

Phonologically conditioned affix order in Yidjɪn

An argument for stratality

Marie-Luise Popp

November 4, 2025

Abstract

In this paper, I argue that phonologically conditioned affix order patterns in Yidjɪn are a) phonologically optimizing, b) driven by independently motivated constraints on the foot structure, and c) overwriting an affix string produced in the morphology, thereby proving that entire affixes can be reordered in the phonology, contra [Paster \(2009\)](#). I argue that the phonology does not have global access to morphological information but only temporary access to a specific piece of morphological information: morphological color. More specifically, I illustrate that phonologically conditioned affix order applies strictly within a stratum, thus fully matching the predictions by Stratal OT.

Keywords: affix order, Stratal OT, morphology–phonology interface

1 Introduction

In Yidjɪn (yidi1250, Pama-Nyungan, Australia), the relative order of the GOING suffix *-li/-ri* and the COM/CAUS *-ŋa* depends on the syllable count of the root. With bisyllabic roots, the COM/CAUS follows the GOING suffix, while the order is reversed with trisyllabic roots, as illustrated in (1). [Dixon \(1977\)](#) argues that this is a consequence of the fact that *-li/-ri* lengthens the preceding syllable. In both examples, the placement of the affixes ensures that it is the head an iambic foot that is lengthened (the second syllable in (1-a), the fourth syllable in (1-b)).

- (1) Placement of the GOING and COM/CAUS suffix ([Dixon 1977](#): 228)
- a. magi-ri-ŋa-l-ŋ
[ma.gi:.ɹi.ŋa)]
climb.up-GOING-COM/CAUS-CL:L-PRS
 - b. maŋinda-ŋa-li-ŋ
[ma.ŋin.da.ŋa:.liŋ]
walk.up-COM/CAUS-GOING-PRS

With respect to phonologically conditioned affix order (PCAO), [Paster \(2006, 2009\)](#) contrasts the predictions of two approaches: *Subcategorization* approaches assume that certain morphemes come with a phonological condition on affixation, encoded in a *subcategorization frame*. These conditions can be phonologically arbitrary, and are checked in the morphology, which distinguishes this approach from P » M approaches. The latter assumes that phonological rules or constraints can make reference to morphological structure to the effect that entire morphemes can be reordered for reasons of phonological well-formedness, a phenomenon

Paster (2009) refers to as ‘true PCAO’. This approach predicts that PCAO should always be phonologically optimizing, and driven by independently motivated phonological principles. In addition, ‘true PCAO’ is expected to be able to violate other conditions on affix order, such as scope (Rice 2000) or morphotactic considerations. Paster (2009) further argues that no convincing instance of PCAO has yet been reported.

In this paper, I argue that the PCAO pattern in (1) confirms the predictions that Paster (2006, 2009) attributes to ‘true PCAO’ where phonology has access to morphological structure (P»M). I argue that there is independent evidence that an affix string produced in the morphology is overwritten due to general and independently motivated constraints on the well-formedness of feet. While the analysis I put forward falls within the scope of P»M approaches, I argue that phonology does not have *global* access to morphology. Rather, I demonstrate that PCAO in Yidj applies only within a restricted morphological domain, thus matching the predictions of Stratal Optimality Theory, see also Popp and Trommer (2025). Moreover, PCAO follows automatically from the interaction of two independently motivated constraints, the general anti-metathesis constraint LINEARITY-IO and CONTIGUITY-□, which ensures linear adjacency of elements belonging to the same morpheme. In short, the analysis requires only *temporary* access to a rather specific piece of morphological information: morphological color.

This paper is structured as follows. I introduce the general morphological makeup including morphological and phonological properties of different affix groups in Section 2. In Section 3, I suggest an analysis in Optimality Theory that accounts for the observation that PCAO overwrites an affix order produced in the morphology due to the interaction of constraints on foot structure. Section 4 discusses asymmetries between derivational and inflectional affixes in final syllable shortening, thus providing evidence for a stratal architecture. Section 5 concludes.

2 The phonology and morphology of suffixes in Yidj

Verbs in Yidj are entirely suffixing. An overview of the morphological structure is given in Table 1. Dixon (1977) distinguishes three different groups of suffixes, based on their morphological and phonological properties.¹

Slot 1	Slot 2	Slot 3	Slot 4	Slot 5	Slot 6	Slot 7		
root	derivational affixes (-n/-l/-ɿ)			class marker	inflection	clitics		
	COM/CAUS	-ŋa/-maŋa/-maŋa -l		-n/-l/-ɿ	PST	-pu	NOW -(a)la SELF -di	
	INTRANS	-ɿi -n			NPST	-ŋ/-l/-ɿ		
	GOING	-ŋali/-:li/-:ɿi -n			IMP	-[APPROX]		
	COMING	-ŋada/-:da/-:da -n			PURP	-na		
					DAT.SUBORD	-punda		
					CAUS.SUBORD	-pum		
					SUBORD	-fi		

Table 1: Affixes in Yidj (based on Dixon 1977: 210)

Each verbal form may take up to three derivational affixes, which occupy slots 2–4 in Table 1. The derivational affixes include the valency-increasing suffix COM/CAUS, the suffix INTRANS, as well as two aspectual affixes GOING and COMING. The valency-increasing suffix COM/CAUS

¹Throughout this paper, I follow Round (2017) in transcribing the voiced palatal stop with *j* rather than *ɟ* used by Dixon (1977).

has two core functions. First, it serves as a comitative applicative. As such, it has two syntactic effects: first, it turns an oblique argument into a core argument, like *bupa* ‘woman’ in (2), which carries comitative case in the absence of the COM/CAUS verbal suffix in (2-a), but zero-marked ABS case in the presence of the COM/CAUS *-ŋa* in (2-b). Second, the predicate becomes transitive. Consequently, the agent argument bears absolutive case in the absence of a COM/CAUS in (2-a), but ergative case when the verb comes with the COM/CAUS suffix in (2-b).

(2) COM/CAUS as a comitative applicative (Dixon 1977:293)

- a. Wagu:ja-∅ bupa:-y gali-ŋ.
man-ABS woman-COM/CAUS walk-NPST
- b. Wagu:ja-ŋgu bupa- gali:-ŋa-l
man-ERG woman-ABS walk-COM/CAUS-CL:L
‘The man is going with the woman.’

As a causative, the COM/CAUS turns the sole argument of an intransitive verb into an internal argument of a transitive verb. Hence, the NOM-marked 1SG pronoun *ŋayu* in (3-a) appears in ACC case in the presence of the COM/CAUS suffix in (3-b). Additionally, the COM/CAUS suffix introduces a new external argument carrying ERG case.

(3) COM/CAUS as a causative applicative (Dixon 1977:312)

- a. ŋayu warŋgi:-ŋ.
1SG.NOM turn.around-PST
‘I turned around.’
- b. ŋaŋaŋ guŋunu-ŋgu warŋgi-ŋa-l-ŋu.
1SG.ACC wind-ERG turn.around-COM/CAUS-CL:L-PST
‘The wind turned me around.’

The suffix INTRANS *-ji* has variety of functions. It may function as a valency-decreasing reflexive or antipassive marker. In these functions, it combines with transitive predicates and produces intransitive predicates. It may also appear with a transitive root to indicate that the agent argument lacks volitional control over the action, though in this case, the predicate remains transitive. Additionally, it can be applied to both transitive and intransitive verbs, maintaining the original valency, to indicate that the action is either an ongoing and/or continuous action. An example of the INTRANS suffix in its antipassive function is given in (4). Crucially, the INTRANS suffix turns a transitive root with both an ergative and an absolutive argument in (4-a) into an intransitive predicate with the agent argument marked with ABS case and the theme argument marked with OBL case in (4-b).

(4) INTRANS as an antipassive

- a. Wagu:ja-ŋgu guda:ga-∅ wawa:-l.
man-ERG dog-ABS see-CL:L
- b. Wagu:ja-∅ wawa:-ji-ŋu gudaga-nda
man-ABS see-INTRANS-PST dog-DAT
‘The man saw the dog.’

All combinations of derivational affixes are attested. However, the aspectual derivational affixes are mutually exclusive, as the former marks direction towards the center of speech, while the latter markers direction away from the center of speech. Table 1 further shows that some of the derivational affixes have different allomorphs. The choice of the allomorph is not phono-

logically conditioned. Rather, the surface form depends on the verb class of the base. In Yidj, each verb root belongs to one of three inflection classes. The inflection classes are labeled *n*, *l* and *ɿ* based on the final consonant that they entail. The class markers are glossed as CL:N, CL:L and CL:R, respectively and may be omitted in certain phonological environments. A property that distinguishes derivational affixes from other suffixes is the fact that the derivational suffixes also entail a class marker, which is spelled out in slot 5 in Table 1. Specifically, the COM/CAUS brings the class marker *-l*, while all other derivational affixes entail the class marker *-n*. Inflectional affixes, on the other hand, do not have class-dependent allomorphs. Moreover, inflectional affixes are obligatory and mutually exclusive, while derivational affixes are optional. In short, each verbal form in Yidj will occur with exactly one inflectional affix and 0–3 derivational affixes. Dixon (1977) further lists clitic-like suffixes, which are given in slot 7 in Table 1. I will not discuss the clitic-like suffix in considerable detail here, since they do not participate in the phonological processes that are relevant for discussion.

Table 1 further reveals that some allomorphs of derivational affixes have the property of lengthening the preceding syllable. The examples in (5-a,b) show that the INTRANS suffix lengthens the preceding syllable, independent of the syllable count of the root. For the GOING and the COMING suffixes, only the *l* and *ɿ* class allomorphs lengthen the preceding syllable, as exemplified in (5-c).

(5) Vowel length through affixation (Dixon 1977: 218, 223)

- a. wuɲaba-ɿi-n
[wu.ɲa.ba.ɿin]
hunt.for-INTRANS-CL:N
- b. buga-ɿi-n
[bu.ga.ɿin]
eat-INTRANS-CL:N
- c. wawa-li-n
[wa.wa.lin]
see-GOING-CL:N

Dixon (1977) observes that the relative order of prelengthening allomorphs of GOING and COMING and the COM/CAUS suffix depends on the number of syllables of the root. In (6-a), the root is the bisyllabic, *l*-class root *magi*. In this case, the *l*-class allomorph of the GOING affix is selected, which is the prelengthening suffix *-ɿi*. In combination with the monosyllabic COM/CAUS affix *ɲa*, the GOING suffix precedes the COM/CAUS, thus lengthening the second syllable of the root. In (6-b), however, the root is trisyllabic. In this case, the GOING suffix follows the COM/CAUS and length goes to the COM/CAUS affix and hence, the fourth syllable of the word.²

(6) Placement of the GOING and COM/CAUS suffix (Dixon 1977: 228)

- a. magi-ri-ɲa-l-ɲ
[ma.gi.ɿi.ɲal])
climb.up-GOING-COM/CAUS-CL:L-PRS
- b. maɟinda-ɲa-li-ɲ
[ma.ɟin.da.ɲa.lin]
walk.up-COM/CAUS-GOING-PRS

²Note that Dixon (1977: 228) states that: ‘With a trisyllabic root, comitative must be directly suffixed to the root [...]’ without explicitly providing empirical evidence that the opposite affix order *maɟinda-li-ɲa-ɲ* is ungrammatical.

Dixon (1977) makes the following empirical generalization: the COM/CAUS and the prelengthening aspectual suffixes always surface in the order that makes the inherent length of the aspectual allomorph surface on an even-numbered syllable. Hence, the relative order of these affixes in Yidj is an instance of phonologically conditioned affix order (henceforth PCAO). In this respect, Paster (2006, 2009) argues that PCAO exists only in the form of phonological subcategorization frames. Subcategorization frames are item-specific conditions on affixation evaluated in the morphology, which can be phonologically arbitrary. Paster (2009) further doubts the existence of a phenomenon Paster (2006, 2009) calls ‘true PCAO’ – reordering of entire affixes in the phonology. Paster (2009) highlights that ‘true PCAO’ makes the predictions that PCAO may result in orders that violate other principles of affix order (e.g. scope, morphology), and that PCAO should always be phonologically optimizing. Paster (2009) further shows that most cases of PCAO do not necessarily fulfil these predictions and can be reanalysed using subcategorization frames. This is a welcomed result given that ‘true’ PCAO requires access to morphological structure by phonological rules (short: P»M). In the remainder of the paper, I will argue that there is good evidence that PCAO in Yidj instantiates phonologically optimizing reordering of an affix string that was produced in the morphology. I will further demonstrate that PCAO and access to morphological structure is limited by strata, rather than global.

3 A formal analysis of PCAO

In the previous section, we saw that derivational affixes are different from other suffixes in affecting the inflection class of the verbal complex. This difference arises most likely from the fact that derivational affixes are derived from independent verb roots. In this section, I will show that exactly this property provides evidence that the morphological component produces an affix order that is later overwritten for reasons of phonological optimization.

Let us now take a deeper look at the relevant PCAO examples, which are enriched with information on the inflection classes in (7). Crucially, the index $\rightarrow X$ indicates that a morpheme creates an inflection class X, whereas the $\textcircled{Y}$ indicates that a marker is the allomorph of inflection class Y. With respect to affix order and allomorphy, examples (7-a-b) are transparent: $:-li/-ri$ is the allomorph appearing after CL:L-roots, like *magi* in (7-a), and produces a CL:N complex, which triggers the CL:N allomorph of the following COM/CAUS $-\eta a$. In (7-b), the trisyllabic root is of inflection class CL:N, explaining the CL:N allomorph $-\eta a$ of the following COM/CAUS suffix, which derives a CL:L-base, to which the CL:L-allomorph of the following GOING suffix attaches. Hence, all allomorphs in (7-a-b) are inward-sensitive to the preceding morph. Example (7-c), however, shows the same affix order and the same allomorphs as (7-a) despite having a root of inflection class CL:N. Crucially, the GOING affix surface in its CL:L allomorph although the preceding root is CL:N.

(7) Placement of the GOING and COM/CAUS suffix (Dixon 1977: 228, 516)

- a. $magi_{\rightarrow L}:-ri_{\textcircled{L}}, \rightarrow N-\eta a_{\textcircled{N}}, \rightarrow L-l$
 $[ma.gi:.i.\eta a.l]$
 climb.up-GOING-COM/CAUS-CL:L
- b. $ma\grave{z}inda_{\rightarrow N}-\eta a_{\textcircled{N}}, \rightarrow L:-li_{\textcircled{L}}, \rightarrow N-\eta$
 $[ma.\grave{z}in.da.\eta a:.lin]$
 walk.up-COM/CAUS-GOING-PRS

- c. $\text{ɟuŋga} \rightarrow \text{N}^- \text{:ri} \text{①}, \rightarrow \text{N}^- \text{ŋa} \text{②}, \rightarrow \text{L}^- \text{l-ŋ}$
 $[\text{ɟuŋ}.\text{ga}:\text{:i.ŋal}]$
 run-GOING-COM/CAUS-CL:L-PRS
- d. $\text{ɟuŋga} \rightarrow \text{N}^- \text{ŋa} \text{②}, \rightarrow \text{L}^- \text{:ri} \text{①}, \rightarrow \text{N}$

As demonstrated above, the example in (7-c) is morphologically opaque: the allomorph of the GOING suffix seems to be outward-sensitive to the COM/CAUS, a phenomenon claimed not to exist (Bobaljik 2000, Paster 2006, Embick 2010). In fact, allomorph selection is only seemingly outward-sensitive. The pattern suggests that the GOING affix must have linearly followed a CL:L-triggering morph – the COM/CAUS – at some point of the morphological derivation in order to explain the surfacing allomorphs. This hypothetical form, where allomorph selection is strictly inward-sensitive, is given in (7-d). I will argue that the intermediate form in (7-d) is later overwritten in the phonology. Thus, Yidjɪ fulfils one of the predictions Paster (2009) ascribes to P»M models: PCAO produces an affix string that disregards other – in this case morphological – principles. Paster (2009) further argues that PCAO as phonological reordering should always be phonologically optimizing, i.e. it should be driven by interperpendently motivated phonological constraints. In the following, I will argue that the alternation in the order of affixes in (7-a-c) follows from general constraints on foot structure. Concretely, I assume that the prelengthening behaviour of certain affixes stems from a floating mora, resulting in the form in (8). This assumption seems rather natural given that Dixon (1977) discusses the lexical verbs *gali* ‘to go’ and *gada* ‘to come’ to be the origin of the GOING and COMING suffixes. While the bisyllabicity has retained in the CL:N allomorph, the CL:L and CL:R allomorphs have been shortened to monosyllabic *li*, with the floating mora being a relict of the deleted first syllable.

$$(8) \quad \begin{array}{cc} \mu & \mu \\ | & | \\ \text{l} & \text{i} \end{array}$$

There is consensus that Yidjɪ builds binary, iambic feet from the left (Dixon 1977, Bown et al. 2013, Round 2017). In an Optimality-Theoretic analysis (Prince and Smolensky 1993), this is modeled with the standardly assumed constraints PARSESYLLABLE, FOOTBINARITY and ALIGN(FT,L,PRWD,L). Bown et al. (2013) further show that feet align with a L+H* pitch accent. With respect to vowel length, I assume a constraint IAMB, which penalizes each foot, whose first syllable (= the non-head syllable) is linked to two moras. These constraints cause the floating mora of the prelengthening affixes to associate to the right syllable of a binary foot. To model the observation that the entire morpheme is reordered along with its mora, I assume the general anti-metathesis constraint LINEARITY-IO which penalizes the metathesis of segments. In addition, I assume the constraint CONTIGUITY-□ (Trommer 2023), which makes the crucial assumption that morphological structure is represented in the phonological representation as *morphological color*, following the ideas by van Oostendorp (2006), Revithiadou (2007), Trommer (2015), Paschen (2018), Zaleska (2018), Trommer (2023). Thus, each morpheme has its own color, which can be referenced by phonological operations, in parallel to features or segments. CONTIGUITY-□ ensures that segments of the same color form a linearly consequent string, thus preventing the separation of morphemes by other elements. The interplay of these two independently motivated constraints ensures that morphemes are metathesised as a whole unit. In other terms, high-ranked CONTIGUITY-□ transforms segment metathesis into morpheme metathesis. Crucially, the CONTIGUITY-□ is independently needed to prevent infixation. The constraint ALTERNATION also makes reference to morphological

color by penalizing the association of floating elements to other elements of the same morpheme. Finally, I assume a constraint MAX_μ which protects moras in the input from deletion. The constraints adopted in this analysis are listed in (9).

- (9)
1. **PARSESYLLABLE:**
Assign one * for each segment in the input that is not dominated by a Foot head in the output.
 2. **FOOTBINARITY (FTBIN):**
Assign one * to every foot that does not contain exactly two syllables.
 3. **ALIGN(FT,L,PRWD,L):**
Assign one * for each syllable between the left edge of a foot and the left edge of a prosodic word.
 4. **IAMB:** Assign a * for each vowel associated with two morae which is not the head of the foot.
 5. **LINEARITY-IO:**
Assign a * for each segment A which precedes segment B in the input, iff A and B have correspondents in the output and the correspondent of A follows the correspondent of B in the output.
 6. MAX_μ :
Assign a * for each μ in the input that has no correspondent in the output.
 7. **ALTERNATION:** (van Oostendorp 2007)
Assign a * for every epenthetic association line between tautomorphic elements.
 8. **CONTIGUITY-□:** adapted from Trommer (2023)
For each pair of morphologically adjacent phonological elements (E_1, E_2) of morphological color C_1 , count a violation for every element of color C_2 that intervenes between E_1 and E_2 .

Let me illustrate how the interplay of these constraints derives and explains PCAO in Yidjip with the tableau in (10). As mentioned above, there is evidence from allomorphy that the underlying form of (7-c) is the affix string *junga-ŋa-ri*. The underlying form, which corresponds to candidate b. in (10), cannot become optimal, since the floating mora associates to the preceding non-head syllable, thus causing a fatal violation of IAMB. Candidate c. prevents a violation of IAMB by deleting the floating mora but fails due to a violation of MAX_μ . Candidate d. preserves the underlying order of morphemes, and does not violate IAMB since the lengthened syllable is the head of a monosyllabic foot. However, this candidate violates FTBIN, just like candidate e., where the lengthened syllable is part of a trisyllabic foot. Candidate f. violates PARSESYLLABLE since two syllables are not parsed into a foot. In candidate g., the floating mora self-associates to a vowel of its own morpheme, thus causing a violation of ALTERNATION. Candidate h. shows the effect of CONTIGUITY-□: the segmental part of the GOING suffix is in its underlying position, while its floating mora associates to the head of first foot. This structure violates neither IAMB nor LINEARITY-IO, since no segments have been metathesised. However, high-ranked CONTIGUITY-□ is fatally violated, since elements of the same morpheme are separated by elements of a different morph (the COM/CAUS *ŋa*). Candidate a. becomes the optimal candidate, which avoids a violation of CONTIGUITY-□ by metathesising the entire morpheme.

(10) Derivation of (7-c)

/ɸuŋ.ga-ŋa:ri/	MAX _μ	CONTIGUITY-□	ALTERNATION	FTBIN	PARSESYLLABLE	ALIGN(Ft.L,PrWd.L)	IAMB	LINEARITY-IO
a. 𐌲𐌿𐌵 (ɸuŋ.ga:).(ri.ŋa)						**		**
b. (ɸuŋ.ga).(ŋa:ri)						**	*!	
c. (ɸuŋ.ga).(ŋa.ri)	*!					**		
d. (ɸuŋ.ga).(ŋa:ri)				*!	*	**		
e. (ɸuŋ.ga.ŋa:ri)				*!	*			
f. (ɸuŋ.ga:).ŋa:ri					*!*			
g. (ɸuŋ.ga).(ŋa.ri:)			*!			**		
h. (ɸuŋ.ga:).(ŋa.ri)		*!				**		

In contrast to (7-c), we cannot deduce the underlying order of morphemes in (7-b). Hence, it cannot be evaluated whether LINEARITY-IO is violated or not. To this end, I list both possible orderings in the input, and note potential violations (*) of LINEARITY-IO in (11) to demonstrate that the candidate a. becomes optimal even if reordering had taken place. Candidate b. with the reversed order of suffixes fatally violates IAMB. Candidate c. satisfies IAMB, but violates ALTERNATION since the mora has self-associated to the vowel of the same morphological color. Candidate d. avoids a violation of IAMB by building a trisyllabic foot, which causes a violation of FTBIN. Candidate e., by contrast, satisfies IAMB and FTBIN by infixing prelengthening suffix *-ri* into the root. However, this causes a fatal violation of CONTIGUITY-□.

(11) Derivation of (7-b)

/ma.ɸinda-ŋa:li/ or /ma.ɸinda:li-ŋa/	MAX _μ	CONTIGUITY-□	ALTERNATION	FTBIN	PARSESYLLABLE	ALIGN(Ft.L,PrWd.L)	IAMB	LINEARITY-IO
a. 𐌲𐌿𐌵 (ma.ɸin).(da.ŋa:).li					*	**		(**)
b. (ma.ɸin).(da:li).ŋa					*	**	*!	
c. (ma.ɸin).(da.li:).ŋa			*!		*	**		
d. (ma.ɸin.da:).(li.ŋa)				*!		***		
e. (ma.ɸi:n).(li.da).ŋa		*!			*	**		**

In the analysis above, PCAO is clearly phonologically optimizing: affix metathesis takes place in order to satisfy independently motivated constraints on foot structure. It needs to be emphasized that the data is compatible not only with an optimization approach, but also with subcategorization frames. Specifically, it is empirically conceivable that the prelengthening allomorphs of the GOING suffix have a subcategorization frame stating that it needs to attach to the right of a foot. However, it is crucial that the subcategorization requirements are checked in a post-morphological module to account for the fact that the observed order violates the morphologically-conditioned ordering. I take these findings to assume the following architecture:

morphology:	concatenation of derivational affixes → allomorph selection
phonology:	footing of syllables → PCAO

Table 2: Assumed architecture of the morpho-phonology interface

I have demonstrated that PCAO in Yidjip requires access by phonological rules to minimal morphological information: morphological color. In the literature, it is argued most prominently by Embick (2010) that a global interaction of phonology and morphology should be banned. One reason for this scepticism comes from Paster (2006, 2009), who argues that a model where phonology has unlimited access to morphological structure would overgenerate by predicting unrestricted non-local phonological requirements on morphology. There is, however, independent evidence in Yidjip that phonological access to morphological color is not global, but rather limited by strata. As mentioned above, the INTRANS suffix *:ji* is also prelengthening, as shown in (12-a-b). Example (12-a) provides additional evidence for CONTIGUITY- $\square$. Crucially, *:ji* does not infix into the root. Rather, the mora associates to the non-head in this case, as shown in the tableau in (13), where candidate b. becomes optimal despite violating IAMB. Importantly, a subcategorization approach would require an assumption similar to CONTIGUITY- $\square$ to exclude candidate c. Hence, a P»M approach requires exactly the same assumption and information as a subcategorization approach, but has the benefit that it naturally explains the class of affixes involved in PCAO by connecting the phenomenon to foot structure.

(12) Prelengthening INTRANS *-:ji* (Dixon 1977:218)

- a. wuŋaba:*:ji*-n
[wu.ŋa.ba:*:jin*]
hunt.for-INTRANS-CL:N
- b. buga:*:ji*-n
[bu.ga:*:jin*]
eat-INTRANS-CL:N

(13) Derivation of (12-a)

/wu.ŋa.ba: <i>:ji</i> /		MAX _μ	CONTIGUITY- $\square$	ALTERNATION	FTBIN	PARSESYLLABLE	ALIGN(FT,L,PrWd,L)	IAMB	LINEARITY-IO
a.	(wu.ŋa)(ba: <i>:ji</i>)			*!			**	*	
b.	 (wu.ŋa)(ba: <i>:ji</i>)						**	*	
c.	(wu.ŋa:)(<i>:ji</i> .ba)		*!				**		
d.	(wu.ŋa.ba:). <i>:ji</i>				*!	*			

In (14), the prelengthening INTRANS *-:ji* is combined with another monosyllabic suffix, the PURP marker *na*. Crucially, the relative order of these two affixes does *not* depend on the syllable count of the root, compare (14-a) and (14-b). Rather, the INTRANS invariantly precedes the PURP *na*. In (14-a), this order is phonologically optimal: the floating mora of the INTRANS associates to the head of the preceding binary foot. In (14-b), however, the mora seems to self-associate. We would have expected reordering of the two suffixes since the phonological structure is similar to (7-b), where reordering would have allowed the floating mora to associate to the head of the preceding foot, see (14-c). Moreover, vowel length does not go to the non-head of the foot either, as in (14-d), although (12-a) clearly shows that this is a possible outcome. Note that Dixon (1977) does not explicitly state the ungrammaticality of (14-c-d), but lists (14-b)

only.

- (14) Relative order of INTRANS *-ji* and PURP *na* (Dixon 1977: 218)
- a. buga-*ji*-na
[(bu.ga:).(ji.na)]
eat-INTRANS-PURP
 - b. wuŋaba-*ji*-na
[(wu.ŋa).(ba.ji:).na]
hunt.for-INTRANS-PURP
 - c. *[(wu.ŋa).(ba.na:).ji] (not attested)
 - d. *[(wu.ŋa).(ba.ji).na] (not attested)

In the following, I will argue that PCAO does not apply in (14-b) since the PURP *-na* is concatenated at a later stratum and hence, only *after* PCAO has applied. Evidence for this claim comes from final syllable shortening, an independent phonological process in which inflectional affixes like the PURP *-na* behave differently than derivational affixes (the prelengthening affixes or the COM/CAUS). Importantly, PCAO cannot apply again at a later stratum because *bracket erasure* bounds access to morphological color – indispensable information for PCAO – to a stratum, thus limiting the predictive power of the system.

4 Affix asymmetries in final syllable shortening: Evidence for strata

The previous section has uncovered a discrepancy between derivational and inflectional affixes: the relative order of prelengthening suffixes and *derivational* monosyllabic suffix *-ŋa* depends on phonological properties of the root, whereas *inflectional* monosyllabic affixes are strictly external to prelengthening suffixes. In this section, I discuss a distinct phonological process final syllable shortening, which reveals another asymmetry between derivational and inflectional affixes. The non-uniform behaviour of inflectional affixes provides evidence that they attach at a later morphophonological domain, hence counter-feeding PCAO.

Yidjŋ has another phonological process that makes reference to syllable count. The vowel of the final syllable is deleted if non-deletion would result in a word with an odd number of syllables. In general, final syllable deletion follows the rule in (15).

- (15) Final syllable deletion
CV(C).CV(C).CV $\Rightarrow$ CV(C).CV:(C)C

If a word ends up with an odd number of syllables, the vowel of the final syllable is deleted, the onset of the final syllable becomes the coda of the preceding consonant, and the vowel of the preceding syllable is lengthened. A simple example of final syllable shortening is illustrated in (16). In (16-a), the trisyllabic root *gindanu* combines with the ergative case marker *ŋgu*. The resulting complex word has four syllables, which fall into two binary feet. In absolutive contexts, however, the root does not carry any affixes, and remains trisyllabic. As a result, the final vowel is deleted, and the onset of the final syllable /n/ becomes the coda of the preceding syllable.

- (16) Word-final deletion (Dixon 1977: 57)
- a. /gindanu-ŋgu/
[(gin.da)(nuŋ.gu)]

- ‘moon-ERG’
 - b. /gindanu/
 - [(gin.da:n)]
 - ‘moon.ABS’

Final syllable deletion affects both roots and suffixes, with exceptional non-participating morphemes in both groups. As an example, none of the prelengthening suffixes undergoes final syllable shortening, as shown in (17). Although the combination of prelengthening suffixes and bisyllabic roots yields words with three syllables in total, the final vowel is not deleted, see (17-b,d). While the ungrammaticality of (17-b) could be explained by the fact that /ʃ/ is not a licit coda, the ungrammaticality of (17-d) is unexpected. Note that the ungrammatical forms in (17-b,d) are not explicitly listed, but inferred from (17-a,c) and the fact that the process is non-optional. Dixon (1977) concludes that the participation in final syllable shortening is an idiosyncratic property of the affixes, see also Round (2017) for a closer discussion.

- (17) No final shortening with prelengthening affixes (Dixon 1977: 218)
- a. buga-:ʃi
 - [(bu.ga:).ʃi]
 - eat-INTRANS
 - b. *[(bu.ga:ʃ)] (not attested)
 - c. wawa-:li
 - [(wa.wa:).li]
 - see-GOING
 - d. *[(wa.wa:l)] (not attested)

In general, final syllable deletion only applies when shortening produces licit codae. This is illustrated in (18). The COM/CAUS shortens, as in (18-a). After a nasal-final root, however, deletion of the vowel would produce the illicit coda combination /ŋŋ/. It is not possible to delete the suffixes altogether, as in (18-e). The same observation holds for roots, see (18-f). The root *binarŋa* does not shorten to *bina:rŋ*, since /rŋ/ is not a licit coda. Again, it is not possible to delete consonantal root material to produce licit codae, either, as in (18-h).

- (18) Roots and COM/CAUS (Dixon 1977: 55, 349, 478), (Round 2017: 63, 64)
- a. /gada-ŋa/
 - [gada:ŋ]
 - come-COMIT/CAUS
 - b. *[[gadaŋa]]
 - c. /gurip-ŋa/
 - [(gu.ri:p)ŋa]
 - ‘good-COMIT/CAUS’
 - d. *[(gu.ri:pŋ)]
 - e. *[(gu.ri:p)]
 - f. /binarŋa/
 - [bina:rŋa]
 - ‘warn’
 - g. *[(bina:rŋ)]
 - h. *[(bina:r)]

In the previous section, we have seen that monosyllabic derivational affixes behave differently in PCAO than monosyllabic inflectional affixes. Another asymmetry between the two groups

arises in final syllable shortening. The examples in (18) have demonstrated that the COM/CAUS suffix will only shorten if the resulting word ends in a licit coda. The same generalization holds for inflectional affixes. However, consonants of inflectional affixes may be deleted along with their vowels to produce licit codae. This is illustrated in (19). The example in (19-a) involves the inflectional affix *punda*. Final syllable deletion would result in the form in (19-b), which involves an illicit coda */nd/*. As a consequence, a consonant is deleted in (19-a), resulting in a word with two binary feet and a licit coda.³ This process might even cause the deletion of entire affixes, as in (19-d). Hence, the difference between derivational *ŋa* and inflectional affixes in final syllable deletion cannot be reduced to the fact that deleting both the vowel and the consonant of *-ŋa* would delete the entire affix.

- (19) Shortening in inflectional affixes (Dixon 1977: 277), (Round 2017: 64)
- a. /binarŋa-l-punda/
[(bi.nar)(ŋalpu:n)]
'warn-CL-DAT.SUBORD'
 - b. *[(bi.nar)(ŋalpu:nd)]
 - c. *[(bi.nar)(ŋalpu:n).da]
 - d. /wawa-l-pu/
[wawa:l]
'see-CL:L-PST'

In sum, final syllable deletion reveals another asymmetry between derivational *-ŋa* and inflectional affixes: as for the COM/CAUS suffix, the final vowel may be deleted to produce words with binary feet if the resulting word contains a licit coda. With inflectional affixes, however, consonantal material may be deleted alongside to produce a licit coda. To model the fact that phonological processes target the COM/CAUS differently than inflectional affixes, I argue that phonological operations apply in two different derivational steps. Following the ideas of Stratal Optimality Theory (Kiparsky 2000, Bermúdez-Otero 2012, 2018), I assume that words in Yidiñ are divided into two domains. In the morphological component of the first stratum, the *stem level*, derivational affixes like COM/CAUS *ŋa* and the prelengthening affixes are concatenated. Stem level phonology triggers PCAO in the way I sketched in Section 3. Once the phonological computation at stem level is complete, bracket erasure applies and deletes access to morphological color, thus excluding stem level morphology from color-sensitive processes like PCAO at a later stratum. At *word level*, morphology connects the complex of root and derivational affixes with the inflectional affixes. Final syllable deletion then applies as part of the word level phonology. The division of labour across components and modules is illustrated in Table 3.

morphology:	concatenation of root and derivational affixes	<i>stem level</i>
phonology:	PCAO & syllabification	
<i>bracket erasure</i>		
morphology:	concatenation of inflectional affixes	<i>word level</i>
phonology:	final syllable deletion & syllabification	
<i>bracket erasure</i>		

Table 3: Assumed architecture of the morpho-phonology interface
The proposed architecture neatly derives the affix asymmetries both in PCAO and in final syllable deletion. First, inflectional affixes do not participate in PCAO since PCAO applies at

³Again, the ungrammatical examples are not explicitly given but inferred from the discussion in Dixon (1977) and Round (2017).

stem level and hence, *before* inflectional affixes are even concatenated at word level. Bracket erasure, a fundamental axiom of Stratal OT, further deletes access to morphological color by phonological rules at the end of the first stratum. As a consequence, the constraint responsible for PCAO CONTIGUITY- $\square$ can never operate on stem level morphology in the phonology of the word level. Thus, PCAO can ever only apply strictly within a stratum, independent of the constraint ranking. This consequence shows that bracket erasure makes cyclic models less powerful than models in which phonology has global access to morphological information: I suggest that phonological grammars have only *temporary* access to a rather specific piece of morphological information: morphological color. Second, the cyclic make-up of the word in Table 3 can also explain the non-uniformity of affixes in final syllable deletion under the assumption that consonants of derivational affixes are inhibited from deletion since they have already been syllabified at stem level when final syllable deletion applies while consonants of inflectional affixes have not. This idea could be modeled by adopting a constraint that prohibits deletion of syllabified consonants.⁴

5 Conclusion

In this paper, I have argued that Yidjñ exhibits reordering of suffixes for phonological optimization, thus instantiating a case of ‘true PCAO’. I have illustrated that the relative order of the prelengthening suffixes and the derivational COM/CAUS suffix *ŋa* depends on the syllable number of the root, and is driven by universal constraints on the well-formedness of feet. Furthermore, I have provided evidence from allomorph selection that PCAO in Yidjñ overwrites an affix string produced in the morphology, thus fulfilling a clear prediction of P»M approaches. While the analysis I suggest falls into the analytical space of a P»M approach, I strongly argue against a *global* access to morphological structure by phonological constraints. Rather, I propose a cyclic model couched in Stratal Optimality Theory in Section 4, in which access to morphological structure is bounded to a stratal domain. Evidence for this claim comes from asymmetries between derivational and inflectional affixes in PCAO and final syllable deletion. Inflectional suffixes do not participate in PCAO, suggesting that they attach at a later stratum, thus counter-feeding PCAO.

Another core result of the paper is the observation that ‘true PCAO’ follows from an interaction of the anti-metathesis constraint LINEARITY-IO and CONTIGUITY- $\square$, which requires adjacency of elements belonging to the same morpheme. Crucially, CONTIGUITY- $\square$ is also necessary in subcategorization approaches. As a consequence, a cyclic P»M approach makes similar assumptions as subcategorization approaches, but is empirically more adequate in providing a phonologically predictable explanation for the pattern.

Abbreviations

ABS = absolutive, ACC = accusative, CAUS = causative, COM = comitative, DAT = dative, ERG = ergative, GOING = going aspect, CL = inflectional class, IMP = imperative, INTRANS = intransitivizer, NOM = nominative, NPST = nonpast, PRS = present, PST = past, PURP = purposive, SUBORD = subordinate

⁴It is worth noting that the discrepancy between derivational and inflectional affixes in final syllable deletion is fully predicted by the assumptions of Harmonic Layer Theory (Trommer and Zimmermann 2021, Zimmermann and Trommer 2022, Tebay et al. 2025), which suggests that the activity of phonological elements changes gradually from cycle to cycle to the effect that derivationally older elements behave differently.

References

- Bermúdez-Otero R (2012) The architecture of grammar and the division of labour in exponence. In: Trommer J (ed) *The morphology and phonology of exponence*, vol 41, Oxford University Press, Oxford, pp 8–83
- Bermúdez-Otero R (2018) Stratal phonology. In: *The Routledge handbook of phonological theory*, Routledge, London, pp 100–134
- Bobaljik JD (2000) The ins and outs of contextual allomorphy. *University of Maryland working papers in linguistics* 10(1):35–71
- Bowern C, Alpher B, Round E (2013) Yidiny stress, length, and truncation reconsidered. Poster presented at NELS44 at University of Connecticut
- Dixon RM (1977) *A grammar of Yidin*. 19, Cambridge University Press
- Embick D (2010) *Localism versus globalism in morphology and phonology*, vol 60. MIT Press, Cambridge
- Kiparsky P (2000) Opacity and cyclicity. *The Linguistic Review* 17(2-4):351–366
- van Oostendorp M (2006) A theory of morphosyntactic colours. Ms, Meertens Institute, Amsterdam Available under: [http://egg.auf.net/06/docs/Hdt% 20Oostendorp% 20coulours pdf](http://egg.auf.net/06/docs/Hdt%20Oostendorp%20coulours.pdf)
- van Oostendorp M (2007) Derived environment effects and consistency of exponence. In: Blaho S, Bye P, Krämer M (eds) *Freedom of analysis?*, vol 95, pp 123–148
- Paschen L (2018) The interaction of reduplication and segmental mutation: A phonological account. PhD thesis, Universität Leipzig Leipzig
- Paster M (2006) Phonological conditions on affixation. PhD thesis, University of California, Berkeley
- Paster M (2009) Explaining phonological conditions on affixation: Evidence from suppletive allomorphy and affix ordering. *Word structure* 2(1):18–37
- Popp ML, Trommer J (2025) Phonologically conditioned affix order is stratal. Manuscript, <https://ling.auf.net/lingbuzz/008911>
- Prince A, Smolensky P (1993) *Optimality Theory: Constraint interaction in generative grammar*. Tech. rep. RuCCS-TR-2 at Rutgers University
- Revithiadou A (2007) Colored turbid accents and containment: A case study from lexical stress. In: Blaho S, Bye P, Krämer M (eds) *Freedom of analysis*, vol 149, Mouton de Gruyter, Berlin/New York, p 174
- Rice K (2000) *Morpheme Order and Semantic Scope: Word Formation in the Athapaskan Verb*. Cambridge Studies in Linguistics, Cambridge University Press, Cambridge
- Round E (2017) Phonological exceptionality is localized to phonological elements: The argument from learnability and Yidiny word-final deletion. In: Bowern C, Horn L, Zanuttini R (eds) *On looking into words (and beyond): Structures, relations, analyses*, Language Science Press Berlin, pp 59–97
- Tebay SE, Trommer J, Zimmermann E (2025) The OCP across languages: An argument for Harmonic Layer Theory. Manuscript, <https://ling.auf.net/lingbuzz/008916>
- Trommer J (2015) Moraic affixes and morphological colors in Dinka. *Linguistic Inquiry* 46(1):77–112
- Trommer J (2023) The concatenative structure of tonal overwriting. *Linguistic Inquiry* 55(1):95–151
- Trommer J, Zimmermann E (2021) Getting stronger or weaker at every stratum: A new approach to tonal morphophonology. Poster presented at Manchester Phonology Meeting on May 26th, 2021
- Zaleska JK (2018) *Coalescence without coalescence*. PhD thesis, Universität Leipzig

Zimmermann E, Trommer J (2022) Getting weaker across layers: The tonal phonology of Shona without stratal re-ranking. Talk given at the linguistics colloquium at University of Warsaw on June 17, 2022